

CSD101 - Discrete Structures (Discrete Mathematics) Fall 2016

Lecture - 8

Sets

Application of Sets

- Databases
- Data-type or type in computer programming
- Constructing discrete structures
- Finite state machine
- Modeling computing machine
- Representing computational complexity of algorithms

Set

- A **set** is an unordered collection of objects.
- The objects in a set are called the **elements**, or **members**, of the set.
- A set is said to contain its **elements**.

Example:

- **Z** is the set of integers.
- Cities in the Pakistan: {Lahore, Karachi, Islamabad, ... }
- Sets can contain non-related elements: {3, a, red, Gilgit }

Properties:

- Order does not matter
 - {1, 2, 3, 4, 5} is equivalent to {3, 5, 2, 4, 1}
- Sets do not have duplicate elements
 - Consider the list of students in this class
 - It does not make sense to list somebody twice

Set Membership

- **a** is an element of the set **A** , denoted by **$a \in A$** .
- **a** is not an element of the set **A** , denoted by **$a \notin A$** .

Sets (example)

- **Example:**
- **Set D:** Students taking Discrete Mathematics course.
- Assume Ali is taking Discrete Mathematics course and Saad is not taking Discrete Mathematics course.
- Ali $\in D$
- Saad $\notin D$

Sets (example)

- Example:

$V: \{a, e, i, o, u\}$

$a \in V$

$b \notin V$

$I: \{0, 1, 2, \dots, 99\}$

$50 \in I$

$100 \notin I$

$S: \{a, 2, \text{class}\}$

$2 \in S$

$\text{room} \notin S$

Specifying a Set

- Capital letters ($A, B, S\dots$) for sets
- Italic lower-case letter for elements ($a, x, y\dots$)
- Easiest way: list all the elements
 - $A = \{1, 2, 3, 4, 5\}$, Not always feasible!
- May use ellipsis (...): $B = \{0, 1, 2, 3, \dots\}$
- May cause confusion. $C = \{3, 5, 7, \dots\}$. What's next?
- If the set is all odd integers greater than 2, it is 9
- If the set is all prime numbers greater than 2, it is 11

Set Builder

- **Set builder:**

Characterize all elements in the set by stating properties they must have.

- **Example:**

$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$

$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$

$O = \{1, 3, 5, 7, 9\}$

The vertical bar means “such that”

Important Sets

- Set of **natural numbers**
 - $\mathbf{N} = \{1, 2, 3, \dots\}$
- Set of **integers**
 - $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- Set of **positive integers**
 - $\mathbf{Z}^+ = \{1, 2, 3, \dots\}$
- Set of **rational numbers**
 - $\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, \text{ and } q \neq 0\}$
- Set of **real numbers**
 - $\mathbf{R}$

Examples

- $S_1 = \{ \mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R} \}$
 - S_1 has 4 elements, each of which is a set.
- $S_2 = \{x \mid x \in \mathbf{N} \text{ and } \exists k \ k \in \mathbf{N}, x = k^2\}$
 - Set of squares of natural numbers

Equality of Sets

- Let A and B be two sets.
 - A and B are **equal** if and only if they have the same elements, denoted by $A = B$.
 - A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$.

Equality of Sets (examples)

- $\{1,2,3\}$ and $\{3,2,1\}$

$$\{1,2,3\} = \{3,2,1\}$$

- $\mathbf{Z}^+$ and $\{0,1,2,\dots\}$

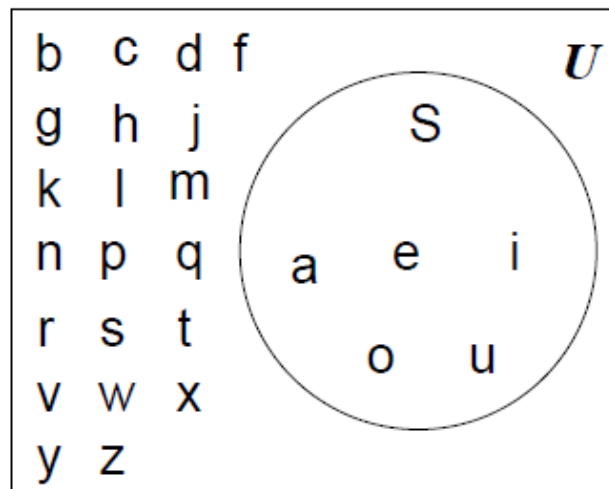
$$\mathbf{Z}^+ \neq \{0,1,2,\dots\}$$

The Universal Set

- **U** is the universal set – the set of all of elements (or the “universe”) from which given any set is drawn
- For the set $\{-2, 0.4, 2\}$, U would be the real numbers
- For the set $\{0, 1, 2\}$, U could be the N , Z , Q , R depending on the context
- For the set of the vowels of the alphabet, U would be all the letters of the alphabet

Venn Diagrams

- Sets can be represented graphically using Venn diagram.



- The box represents the universal set
- Circles represent the set(s)
- Consider set S , which is the set of all vowels in the alphabet
- The individual elements are usually not written in a Venn diagram

Empty Set (example)

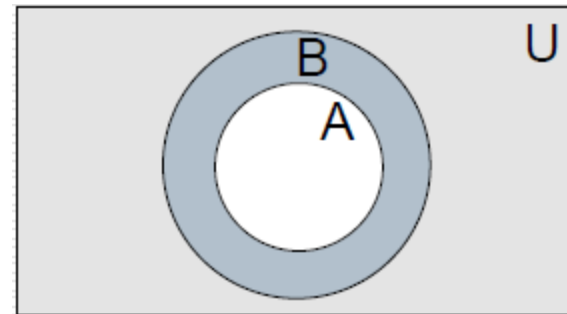
- Example:
- $S = \{x \mid x \in \mathbb{Z}^+ \text{ and } x < 0\}$
 $S = \{\} = \emptyset$
- A set that has no elements called **empty set**, or **null set**.
- $\emptyset$ and $\{\emptyset\}$
 $\emptyset \neq \{\emptyset\}$

Sets Of Sets

- Sets can contain other sets
- $S = \{ \{1\}, \{2\}, \{3\} \}$
- $T = \{ \{1\}, \{\{2\}\}, \{\{\{3\}\}\} \}$
- $V = \{ \{ \{1\}, \{\{2\}\} \}, \{ \{\{3\}\} \}, \{ \{1\}, \{\{2\}\}, \{\{\{3\}\}\} \} \}$
V has only 3 elements!
- Note that $1 \neq \{1\} \neq \{\{1\}\} \neq \{\{\{1\}\}\}$
- They are all different

Subset

- Let A and B be sets.
- A is a **subset** of B if and only if every element of A is also an element of B, denoted by $A \subseteq B$.
- $A \subseteq B$ if and only if $\forall x (x \in A \rightarrow x \in B)$.
- $\forall$ set S,
 $\emptyset \subseteq S$
 $S \subseteq S$



Subset and Equality

- $A \subseteq B, \forall x (x \in A \rightarrow x \in B)$
and
- $B \subseteq A, \forall x (x \in B \rightarrow x \in A)$
then
- $A = B, \forall x (x \in A \leftrightarrow x \in B)$

Subset (example)

- **Q** and **R**

$$\mathbf{Q} \subseteq \mathbf{R}$$

- **N** and **Z**

$$\mathbf{N} \subseteq \mathbf{Z}$$

- $A = \{x \mid x \in \mathbf{Z}^+ \text{ and } x < 10\}$

$$B = \{x \mid x \in \mathbf{Z}^+, x \text{ is even and } x < 10\}$$

$$B \subseteq A$$

Subset

- Show $\forall$ set S , $\emptyset \subseteq S$.

- Proof:

We want to show $\forall x (x \in \emptyset \rightarrow x \in S)$.

- $\emptyset$ contains no element, so $x \in \emptyset$ is false.
- Hypothesis of conditional statement is false, so $x \in \emptyset \rightarrow x \in S$ is true.
- Thus, $\forall x (x \in \emptyset \rightarrow x \in S)$ is true.

Subset

- Show $\forall$ set S , $S \subseteq S$.

- Proof:

We want to show $\forall x (x \in S \rightarrow x \in S)$.

- If $x \in S$ is true, then hypothesis and conclusion of conditional statement are both true and $(x \in S \rightarrow x \in S)$ is true.
- If $x \in S$ is false, then hypothesis and conclusion of conditional statement are both false and $(x \in S \rightarrow x \in S)$ is true.
- Thus, $\forall x (x \in S \rightarrow x \in S)$ is true.

Proper Subset

Let A and B be sets.

- A is a **proper subset** of B if and only if $A \subseteq B$ but $A \neq B$, denoted $A \subset B$.
- $A \subset B$ if and only if $\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$.

Example

- If S is a subset of T , and S is not equal to T , then S is a proper subset of T

Let $T = \{0, 1, 2, 3, 4, 5\}$ and $S = \{1, 2, 3\}$

- S is not equal to T , and S is a subset of T
- Let $Q = \{4, 5, 6\}$. Q is neither a subset of T nor a proper subset of T
- The difference between “subset” and “proper subset” is like the difference between “less than or equal to” and “less than” for numbers

- Is $\emptyset \subseteq \{1,2,3\}$?
- Is $\emptyset \in \{1,2,3\}$?
- Is $\emptyset \subseteq \{\emptyset,1,2,3\}$?
- Is $\emptyset \in \{\emptyset,1,2,3\}$?

- Is $x \in \{x\}$
- Is $\{x\} \subseteq \{x\}$
- Is $\{x\} \in \{x,\{x\}\}$
- Is $\{x\} \subseteq \{x,\{x\}\}$
- Is $\{x\} \in \{x\}$

- Is $\emptyset \subseteq \{1,2,3\}$? Yes!
- Is $\emptyset \in \{1,2,3\}$? No!
- Is $\emptyset \subseteq \{\emptyset,1,2,3\}$? Yes!
- Is $\emptyset \in \{\emptyset,1,2,3\}$? Yes!

- Is $x \in \{x\}$
- Is $\{x\} \subseteq \{x\}$
- Is $\{x\} \in \{x,\{x\}\}$
- Is $\{x\} \subseteq \{x,\{x\}\}$
- Is $\{x\} \in \{x\}$

Size of Sets

- Let S be a set.
- The cardinality of a set is the number of elements in a set S
- **cardinality** of S , denoted by $|S|$.

Example

- Find cardinality of following sets.
- $A = \{x \mid x \in \mathbb{Z}^+, x \text{ is odd and } x < 10\}$
 $A = \{1, 3, 5, 7, 9\}$
 $|A| = 5$
- $B = \emptyset$
 $|B| = 0$
- $C = \{\emptyset\}$
 $|C| = 1$
- **R**
R is infinite.

The Power Set

- Let S be a set.
- The **power set** of S is the set of all subsets of S , denoted by $P(S)$.
- Example:
$$P(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

The Power Set (example)

- What is $P(\{1,2,3\})$?

- Solution:

$$P(\{1,2,3\}) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\} \}$$

- $P(\emptyset) = ?$

- $P(\{\emptyset\}) = ?$

The Cardinality of the Power Set

- Assume A is finite.
- $|P(A)| = ?$

Solution:

- $A = \{a\}$ $P(A) = \{\emptyset, \{a\}\}$ $|P(A)| = 2$
- $A = \{a,b\}$ $P(A) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$ $|P(A)| = 4$
- $A = \{a,b,c\}$
 $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$ $|P(A)| = 8$
- $|P(A)| = 2^{|A|}$

Cartesian Product

Let A and B be sets.

- The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a,b) , where $a \in A$ and $b \in B$.
- $A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$

Cartesian Product (example)

$$A = \{0,1,2\}$$

$$B = \{a,b\}$$

Are $A \times B$ and $B \times A$ equal?

Solution:

$$A \times B = \{(0,a),(0,b),(1,a),(1,b),(2,a),(2,b)\}$$

$$B \times A = \{(a,0),(a,1),(a,2),(b,0),(b,1),(b,2)\}$$

So, $A \times B \neq B \times A$.

The Cardinality of Cartesian Product

Assume A and B are finite.

$$|A \times B| = ?$$

Solution:

- $A = \{a\} \quad B = \{0\}$

$$A \times B = \{(a, 0)\} \quad |A \times B| = 1$$

- $A = \{a, b\} \quad B = \{0\}$

$$A \times B = \{(a, 0), (b, 0)\} \quad |A \times B| = 2$$

- $A = \{a, b\} \quad B = \{0, 1\} \quad A \times B = \{(a, 0), (a, 1), (b, 0), (b, 1)\}$

$$|A \times B| = 4$$

- **$|A \times B| = |A| \cdot |B|$**

Cartesian Product

- Let $A_1, A_2, \dots, A_n$ be sets.
- The **Cartesian product** of $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$, where $a_i \in A_i$ for $i = 1, 2, \dots, n$.
- $A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i, \forall i \in \{1, 2, \dots, n\}\}$

Cartesian Product

$$A = \{a, b\}$$

$$B = \{1\}$$

$$C = \{x, y, z\}$$

$$A \times B \times C = ?$$

Solution:

$$A \times B \times C = \{(a,1,x), (a,1,y), (a,1,z), (b,1,x), (b,1,y), (b,1,z)\}$$

The Cardinality of Cartesian product

Assume A, B and C are finite.

$$|A \times B \times C| = ?$$

Solution:

- $A = \{a\}$ $B = \{0\}$ $C = \{x\}$ $A \times B \times C = \{(a, 0, x)\}$
 $|A \times B \times C| = 1$
- $A = \{a, b\}$ $B = \{0\}$ $C = \{x\}$ $A \times B \times C = \{(a, 0, x), (b, 0, x)\}$
 $|A \times B| = 2$
- $A = \{a, b\}$ $B = \{0, 1\}$ $C = \{x\}$
 $A \times B = \{(a, 0, x), (a, 1, x), (b, 0, x), (b, 1, x)\}$ $|A \times B \times C| = 4$
- **$|A \times B \times C| = |A| \cdot |B| \cdot |C|$**

Ordered n-tuple

- The **ordered n-tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, $\dots$, and a_n as its n th element.

- Example:

(a, b) is an ordered 2-tuple (ordered pair).

Ordered n-tuple (example)

- Assume $c \neq b$.
- Are ordered 3-tuples (a,b,c) and (a,c,b) equal?

Solution:

- $a = a$ but $b \neq c$ and $c \neq b$.
- So, (a,b,c) and (a,c,b) are not equal.

Using Set Notation with Quantifiers

- $\forall x P(x)$ domain: S
- $\forall x \in S (P(x))$
- $\forall x (x \in S \rightarrow P(x))$

Using Set Notation with Quantifiers

- $\exists x P(x)$ domain: S
- $\exists x \in S (P(x))$
- $\exists x (x \in S \wedge P(x))$

Example

- What does the following statement mean?

$$\forall x \in \mathbf{R} (x^2 \geq 0)$$

Example

- What does the following statement mean?

$$\forall x \in \mathbf{R} (x^2 \geq 0)$$

Solution:

- For every real number x , $(x^2 \geq 0)$.
- The square of every real number is nonnegative.

Example

- What does the following statement mean?

$$\exists x \in \mathbf{Z} (x^2 = 1)$$

Example

- What does the following statement mean?

$$\exists x \in \mathbf{Z} (x^2 = 1)$$

Solution:

- There is an integer x such that $x^2 = 1$.
- There is an integer whose square is 1.

Truth Sets of Predicates

- Let P be a predicate and D is a domain.
- The **truth set** of P is the set of elements x in D for which $P(x)$ is true.
- The truth set of $P(x)$ is $\{x \in D \mid P(x)\}$.

Example

- Let $P(x)$ be $|x| = 1$ where the domain is the set of integers.
What is the truth set of $P(x)$?

Example

- Let $P(x)$ be $|x| = 1$ where the domain is the set of integers.
What is the truth set of $P(x)$?

Solution:

The truth set of $P(x)$ is $\{-1, 1\}$.

Example

- Let $R(x)$ be $|x| = x$ where the domain is the set of integers.
What is the truth set of $R(x)$?

Example

- Let $R(x)$ be $|x| = x$ where the domain is the set of integers.
What is the truth set of $R(x)$?

Solution:

The truth set of $R(x)$ is $x \geq 0$.

Example

- Let $Q(x)$ be $x^2 = 2$ where the domain is the set of integers.
What is the truth set of $Q(x)$?

Example

- Let $Q(x)$ be $x^2 = 2$ where the domain is the set of integers.
What is the truth set of $Q(x)$?

Solution:

The truth set of $Q(x)$ is $\emptyset$.

Truth Set of Quantifiers

- $\forall x P(x)$ is true over the domain D if and only if the truth set of P is the set D .
- $\exists x P(x)$ is true over the domain D if and only if the truth set of P is nonempty.

Exercise Questions

Chapter # 2

Topic # 2.1

Question # 1,3,5,6,7,9,12,19,20,23,32,43,44